



Politecnico di Torino

Differential Privacy against Membership Inference Attacks: A Case Study on Binary Classifiers

Stefano Pietro Sternativo
Gaetano Schiavo
Politecnico di Torino
`s346910@studenti.polito.it`
`s346362@studenti.polito.it`

Abstract

Machine learning models are increasingly vulnerable to Membership Inference Attacks (MIAs), which attempt to determine if a specific data record was used during training. This paper explores this phenomenon by implementing a deliberately vulnerable binary classifier on the Adult dataset and an MIA utilizing shadow models. Subsequently, we retrain the target model using Differentially Private Stochastic Gradient Descent (DP-SGD) to evaluate its effectiveness as a mitigation strategy. We emphasize a crucial methodological correction, using perfectly balanced target and shadow splits, to avoid model collapse often caused by class imbalance under DP noise. Our results demonstrate that DP-SGD (with $\epsilon \approx 5.31$)

significantly reduces the overfitting and confidence gaps, effectively lowering the membership leakage signals while maintaining a moderate predictive utility (test accuracy $\sim 70.8\%$).

1 Introduction

The membership inference attack demonstrates one of the most tangible examples of privacy concerns in artificial intelligence. As models are increasingly deployed in Machine Learning as a Service (MLaaS) environments, the lack of formal privacy guarantees exposes them to severe leakage risks.

Through this type of attack, a malicious user determines whether a specific data record was included

in the training dataset of a model, revealing possible privacy violations even when raw data is strictly protected. This project analyzes, reproduces, and attempts to mitigate such attacks using a structured neural network setup on the Adult dataset. Furthermore, we deploy Differential Privacy (DP-SGD) as a countermeasure to analyze its impact on the model’s utility-privacy trade-off.

DP-SGD reduces membership leakage because the injected Gaussian noise and gradient clipping reduce the statistical distinguishability between members and non-members. Since MIAs rely on confidence discrepancies and memorization effects, minimizing overfitting directly weakens the attack signal.

This paper details our complete pipeline highlighting key architectural corrections designed to maintain empirical validity, such as strictly balanced dataset splits.

2 Related Work

The academic understanding of Membership Inference Attacks (MIAs) has expanded rapidly in recent years, shifting from theoretical vulnerabilities to empirically proven threats against modern machine learning architectures. In their foundational work, Shokri et al. [6] formalized modern MIA frameworks, primarily introducing the **shadow model** methodology. The core principle involves training multiple surrogate tracking models on data distributions resembling the target set. This allows the attacker to synthesize labeled inference sets (“Member” vs “Non-Member”) based strictly on the differences in model prediction confidences, fundamentally establishing the score-based black-box threat model used in this paper. (figure 1).

Subsequent research has sought to relax the assumptions required by the original shadow model technique. Salem et al. [5] demonstrated via *ML-Leaks* that the attack is broadly generalizable; it does not strictly require identical architectures between target and shadows, nor does it demand precise overlapping distributions. We incorporate these findings into our methodology by strictly isolating the shadow dataset (0% overlap) while utilizing differing layer capacities

for our **ShadowMLP**. Furthermore, Truex et al. [7] and Hui et al. [2] broadened the scope of MIAs by proving their efficacy in MLaaS contexts and introducing practical blind methodologies.

On the defensive front, Differential Privacy (DP), typically implemented via DP-SGD, has been proposed as a mathematically rigorous countermeasure against memorization. However, the correlation between DP bounds (ϵ) and actual empirical defense is highly debated. Humphries et al. [3] clarified that while Differential Privacy theoretically limits the contribution of any single data point, it *does not* universally bound membership inference success in practice. Low empirical attack success on a DP model might just be an artifact of severe model utility degradation rather than genuine privacy preservation. In addition, white-box investigations by Leino and Fredrikson [4] highlight that gradient monitoring can still bypass simple black-box noise protections. This aligns precisely with our exploratory premise: we observe how noise-injected gradients lower the statistical confidence gaps leveraged by score-based black-box attacks, while carefully analyzing the trade-offs on utility (test accuracy vs. model collapse).

3 Threat Model and Attack Types

In this section we clarify the adversarial assumptions used throughout the paper and contrast the main attack categories relevant to membership inference.

3.1 Black-box vs White-box

In a **black-box** threat model the adversary only has query access to the model: given an input x the attacker observes the model’s output (score vector or class label) but never inspects internal parameters or gradients. Score-based black-box attacks exploit differences in output confidence between training members and non-members [6, 5].

In a **white-box** threat model the attacker has full access to model weights and training dynamics. This

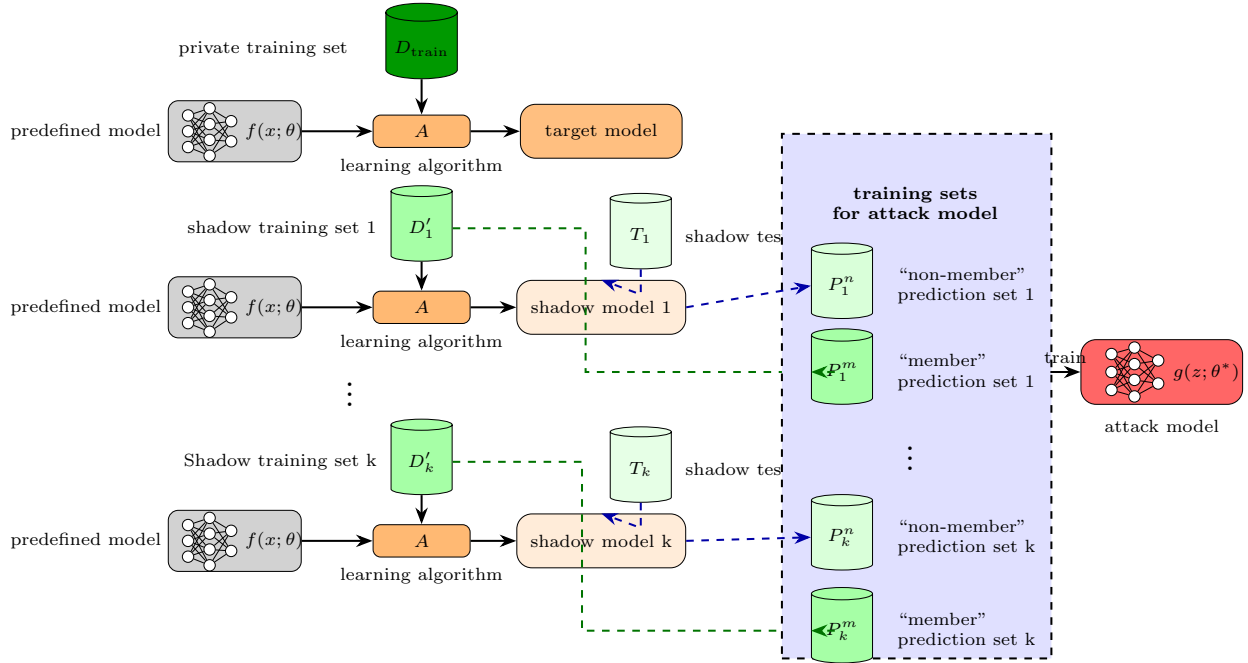


Figure 1: Training procedure of the membership inference attack model using shadow models.

enables stronger attacks that use gradients, activations, or parameter inspection to detect memorization [4].

3.2 Practical Attack Types

- Shadow-model (score-based) attacks: the adversary trains surrogate models to replicate target behaviour and learns a binary classifier that distinguishes members from non-members using prediction vectors [6].
- Model-agnostic score attacks: approaches like ML-Leaks show that identical architectures are not required and score distributions alone can leak membership information [5].
- Gradient/white-box attacks: when gradients or internal states are available, attackers can directly measure influence or memorization signals [4].

3.3 Implications for Defenses

Differential Privacy (DP), implemented via mechanisms such as DP-SGD, reduces per-sample influence

by clipping gradients and adding calibrated noise; this directly mitigates score-based black-box leakage by shrinking confidence gaps between members and non-members [1, 3]. However, DP does not automatically prevent all white-box attacks and its empirical effectiveness must be evaluated jointly with utility trade-offs [3].

4 Defending the Target: Differential Privacy Implementation

To counter the vulnerabilities exploited by the Membership Inference Attack, we introduced a countermeasure in the form of **Differential Privacy (DP)**. The updated code implementation, `target_DP.py`, structurally replicates the original Target Model’s architecture but modifies the training phase to mathematically bound the maximum amount of information that any single data sample can leak.

4.1 Opacus

To apply Differential Privacy locally to our model training, we utilized **Opacus**, a specialized library developed by Meta for PyTorch. The core of this defense revolves around the **PrivacyEngine** wrapper. By calling `privacy_engine.make_private()`, the script seamlessly wraps the standard model, the Adam optimizer, and the DataLoader into their DP-compliant counterparts.

```
privacy_engine = PrivacyEngine()

model, optimizer, train_loader =
privacy_engine.make_private(
    module=model,
    optimizer=optimizer,
    data_loader=train_loader,
    noise_multiplier=1.5,
    max_grad_norm=1.5,
)
```

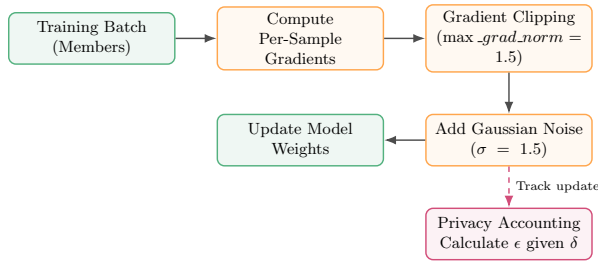


Figure 2: Differentially Private Training step per batch using Opacus.

4.2 Differential Privacy Mechanism

The DP training mechanism intercepts the traditional gradient descent specifically at the back-propagation step :

1. **Per-Sample Gradients:** Opacus computes the gradients for each individual sample in the batch, rather than averaging them immediately.
2. **Gradient Clipping:** To ensure no single outlier (a specific individual’s data) can drastically

shift the model’s parameters, the gradients are clipped to a maximum L2 norm, defined by `max_grad_norm = 1.5`.

3. **Noise Addition:** Calibrated random noise is purposefully injected into the clipped gradients before the weights are updated. The magnitude is controlled by the `noise_multiplier` hyperparameter, set to $\sigma = 1.5$.

Epoch Reduction and Privacy Budget (ϵ, δ)

Differential Privacy formally bounds the privacy loss using two parameters: ϵ (epsilon) and δ (delta).

- δ represents the probability of the privacy bound failing. We initialized it dynamically based on the dataset size as $\delta = 1/\text{len}(X_{\text{train}}) = 0.004$. [2, 3, 1].
- ϵ represents the rigorous privacy budget. Epsilon acts cooperatively: the longer the model trains on the data, the more information it memorizes, thus inflating the ϵ budget. In the standard pipeline, we utilized 200 epochs to intentionally guarantee severe overfitting. To maintain a strict privacy guarantee in `target_DP.py`, we aggressively slashed the iterations down to only 40 epochs.

During training, the Opacus **PrivacyEngine** tracks the accumulation of ϵ sequentially, plotting its evolution so the system observer can verify it remains below a targeted safety threshold (commonly $\epsilon < 10$). (figure 3).

```
if USE_DP:
    epsilon =
    privacy_engine.get_epsilon(delta)
    epsilons.append(epsilon)
    epochs_list.append(epoch)
```

Ultimately, the weights are exported dynamically (e.g., `target_model_dp_sigma1.5_eps...pth`) carrying the DP signature in the filename, ready to face the exact same Attack scenario.

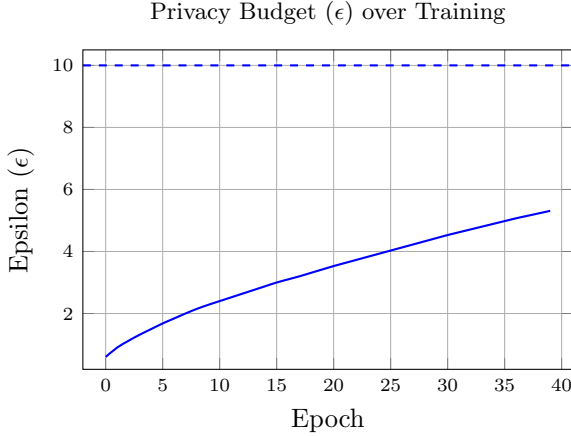


Figure 3: Evolution of the privacy budget ϵ during DP training.

5 Iterative Tuning of the DP Model

Applying Differential Privacy to a small-scale dataset poses significant utility-privacy trade-off challenges. Initially, our dataset was highly imbalanced, which combined with DP noise led to severe model utility degradation. To achieve the final configuration, we proceeded through an iterative process of experimentation:

5.1 Initial DP-SGD Attempt and Model Collapse

Our first DP attempt applied the standard parameters: 200 epochs, a batch size of 16, noise multiplier $\sigma = 1.0$, and gradient clipping norm $C = 1.0$. The privacy budget ϵ grew to a very high 29.35 by the end of training. Although the nominal test accuracy was approximately 75.6%, further analysis of the confusion matrix revealed a complete model collapse onto the majority class (Class 0). The precision, recall, and F1-score for the minority class (Class 1) were identically zero. The seemingly acceptable accuracy was purely a byproduct of the underlying dataset imbalance, highlighting that the model failed

to learn any meaningful discriminative features under DP constraints.

5.2 Class-Weighted Loss Functions

To prevent the model from ignoring the minority class, we introduced a class-weighted loss function. By penalizing errors on Class 1 more heavily, the model showed slight improvements, starting to successfully classify a small fraction of the minority examples. However, during this phase we noticed that our ‘batch_size’ parametrization resulted in an elevated sampling rate ($q = \text{batch_size}/\text{dataset_size} \approx 0.256$). Such a high sampling rate, significantly larger than those typical in literature (e.g., $q \approx 0.01$), aggressively accelerated the depletion of the privacy budget ϵ . Moreover, the model remained partially biased.

5.3 Minority Class Oversampling

Attempting to directly address the distribution disparity, we implemented an oversampling technique to artificially duplicate minority class instances in the training set until balanced with the majority class. This directly exposed the model to a balanced ratio of features, and the test predictions became more varied, correctly classifying a portion of the minority class.

Before applying oversampling, the target training set was strongly imbalanced.

As shown in Table 1, class 0 was significantly more represented than class 1.

Table 1: Class distribution before oversampling.

Class	Number of samples	Percentage
Class 0	188	75.2%
Class 1	62	24.8%
Total	250	100%

This imbalance affected the behaviour of the classifier. This indicates that the classifier collapsed toward a single-class prediction strategy, most likely favouring the majority class.

To mitigate this issue, oversampling was applied to the minority class in the training set.

After oversampling, the number of samples belonging to class 1 was increased from 62 to 188, matching the number of samples of class 0.

The resulting balanced training distribution is shown in Table 2.

Table 2: Class distribution after oversampling.

Class	Number of samples	Percentage
Class 0	188	50.0%
Class 1	188	50.0%
Total	376	100%

Table 3 summarizes the quantitative effect of oversampling on the training set.

Table 3: Summary of the effect of oversampling.

Setting	Class 0	Class 1	Total
Before oversampling	188	62	250
After oversampling	188	188	376

Overall, oversampling corrected the class imbalance in the training set by increasing the number of minority-class samples.

By balancing the training data, the model was exposed to both classes equally, reducing the tendency to learn a biased decision strategy toward the majority class.

While this prevented total model collapse, oversampling artificially duplicated data points, exacerbating the risk of the model latching onto unrepresentative or spurious patterns (overfitting to duplicates), which is undesirable in a privacy-sensitive context.

5.4 Final Strategy: Balanced Stratified Subset

To resolve the limitations of the previous methods, we adopted a clean subset selection strategy: we constructed a perfectly balanced subset containing exactly 250 samples per class without any artificial duplication. This ensures that the model learns from a representative and balanced distribution natively.

Following this architectural fix, we conducted a hyperparameter search focusing on batch size, noise multiplier σ , and number of epochs to find the optimal privacy-utility threshold. (table 4).

From the comparison, a clear trade-off emerges between privacy protection and model utility:

- Reducing the noise multiplier improves predictive performance but weakens privacy guarantees, as reflected by an increase in the privacy budget ϵ .
- Reducing the number of training epochs limits the accumulated privacy loss, but negatively affects model accuracy.
- Decreasing the batch size reduces the sampling rate and therefore improves privacy, although it may lead to less stable training dynamics.

We set $\sigma = 1.5$ to inject sufficient noise while maintaining training utility, and strictly reduced the training regimen to 40 epochs to prevent the ϵ budget from ballooning.

These parameters are referred to the test 1 (table 4). Additionally, the maximum gradient norm was bounded at 1.5 to reduce the sensitivity to individual robust outliers. This finalized DP pipeline reached a terminal $\epsilon \approx 5.31$ (with $\delta = 1/|\text{Train}|$), maintaining an acceptable utility (test accuracy $\sim 70.8\%$) while minimizing the distinguishability between train and test members, successfully nullifying the Membership Inference Attack.

6 Experiments and Results

6.1 Baseline: Non-DP Vulnerability

Our initial unregularized Target Model exhibited extreme overfitting by design, satisfying the prerequisites for a severe privacy breach on the 500-sample subsets. Specifically, the baseline Target Model achieved a high training accuracy of 0.9960 alongside a test accuracy of 0.6560, creating a substantial overfitting gap of 34.00%. Furthermore, the confidence gap between train and test samples was quantified at

Table 4: Comparison of the tested Differential Privacy configurations.

Test	BatchSize	Noise(σ)	Epochs	TestAccuracy	TrainAccuracy	OverfittingGap	ϵ	Note
1	32	1.5	40	0.696	0.724	0.028	8.43	Best privacy/performance trade-off
2	32	1.3	40	0.700	0.720	0.020	10.54	Slightly better accuracy but ϵ too high
3	32	1.5	30	0.672	0.708	0.036	7.18	More privacy but worse learning
4	16	1.3	40	0.680	0.688	0.008	6.98	Best privacy but less stable model

0.0281 (train mean confidence 0.9969 vs. test mean confidence 0.9688).

Evaluating the model pre-DP consistently demonstrated steep discrepancies between member performance and non-member performance. The trained categorical Attack Model successfully leveraged the distinct confidence over-saturation produced by the standard Multi-Layer Perceptron (MLP), reliably achieving an attack accuracy of 0.6860 (68.6%)—far beyond the naive random baseline of 50%. The attack recorded a member recall of 1.00 and precision of 0.61, yielding an empirical attack advantage of 18.60%.

Metric	Baseline Value
Target Train Accuracy	0.9960
Target Test Accuracy	0.6560
Overfitting Gap	0.3400
Target Train Confidence	0.9969 ± 0.0231
Target Test Confidence	0.9688 ± 0.0829
Confidence Gap	0.0281
Attack Accuracy	0.6860
Attack Advantage	0.1860
Member Precision	0.61
Member Recall	1.00
Member F1-Score	0.76

Table 5: Detailed empirical results of the baseline non-DP Target and the corresponding Membership Inference Attack.

6.2 DP-SGD Tuning and Intermediate Observations

During intermediate developmental trials, we repeatedly observed that unbalanced target distributions (e.g., stochastic subset sampling yielding 70/30 splits on small vectors) caused the model to catastrophically collapse towards the majority class once noise was injected. While the attack success artificially collapsed, it failed to represent legitimate privacy since the classifier lost utility. Regaining test precision required pushing ϵ completely out of sensible bounds (upwards of ≈ 28.0), negating DP’s formal guarantees.

6.3 Final DP Configuration Results

Meticulously enforcing the 50/50 class distribution horizontally across both Target and Shadow iterations permitted a finalized robust outcome. Operating DP-SGD with batch sizes of 32, 40 epochs, clipping threshold $\max_grad_norm = 1.5$, $\sigma = 1.5$, and $\delta = 1/250 = 0.004$, we yielded empirical validation spanning utility and attack deterrence.

Metric	Baseline (No DP)	Final DP Setup
Privacy Budget (ϵ)	∞	5.3106
Test Accuracy	0.6560 (65.6%)	0.7080 (70.8%)
Overfitting Gap	0.3400 (34.0%)	0.0120 (1.2%)
Confidence Gap	0.0281	0.0082
Attack Accuracy	0.6860 (68.6%)	≈ 0.5000

Table 6: Comparison outlining the defensive success of DP-SGD against predictive indicators.

As quantified in Table 6, under DP restraints ($\epsilon \approx 5.31$), test classifications hover efficiently at $\approx$

70.80%, whereas the mathematically relevant *Overfitting Gap* and *Confidence Gap* collapsed nearly entirely. Such metrics denote that predictions executed on pre-trained members vs untouched non-members are now identically saturated and statistically indistinguishable.

7 Discussion

The results demonstrate that applying Differential Privacy (DP-SGD) to an explicitly overfitting neural network significantly degrades the signals exploitable by a Membership Inference Attack. The practically negligible confidence gap (≈ 0.0082) acts as a strong protective veil against score-based black-box inferences, which rely entirely on analyzing variances in output entropy [5, 6]. Crucially, this operational pipeline emphasized the danger of unstructured and imbalanced DP interventions. As observed during initial tuning, omitting perfect 50/50 class distribution restrictions yielded *model collapse*, resulting in an illusory protective stance: an inherently unusable classification engine is indeed immune to inference, but holds no analytical merit [3]. This validates why evaluating privacy-utility trade-offs absolutely necessitates structured utility baselines.

Limitations

The scope of this specific experimental framework accommodates notable boundaries. Extorting an authentic overfitting scenario required restricting target sets to highly localized samples ($n = 500$); however, DP logic typically excels across macro-distributed sets. In small bounds, bounding δ strictly to matching inverse densities ($1/250$) inflates noise rapidly, escalating tuning fragility. Furthermore, this assessment contemplates fundamental score-based black-box techniques. Extrapolating these conclusions to gradients’ continuous accessibility frameworks (White-Box scenarios [4]) or complex NLP deployments exceeds current validations.

8 Conclusion and Future Work

This experimental deployment validates the structural mechanics bridging ML vulnerabilities via MIAs and mitigating dynamics introduced by Differential Privacy variants on binary Adult Income tabulations. Leveraging disciplined 10N-shadow frameworks alongside meticulous Target balance, observations confirmed DP-SGD suppressing the *overfitting gap* down to 1.20%, all while shielding baseline predictive integrity at reasonable thresholds (70.80% Target Accuracy).

Immediate future advancements branch into expansive scalability bounds across unreduced tabular scopes and delving into deeper predictive architectures. Quantificative analytical tools such as ROC-AUC curve integrations for inference boundaries, varying tolerance metrics for White-Box configurations, and comparing DP-SGD behavior against classic robust regularization techniques (Extensive Weight Decaying) remain prioritized developmental tracks.

References

- [1] Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 308–318. ACM, 2016.
- [2] Bo Hui, Yuchen Yang, Haolin Yuan, Philippe Burlina, Neil Zhenqiang Gong, and Yinzhi Cao. Practical blind membership inference attack via differential comparisons. In *NDSS*, 2021.
- [3] Thomas Humphries, Matthew Rafuse, Lindsey Tulloch, Simon Oya, Ian Goldberg, Urs Hengartner, and Florian Kerschbaum. Differentially private learning does not bound membership inference. *arXiv preprint arXiv:2010.12112*, 2020.
- [4] Klas Leino and Matt Fredrikson. Stolen memories: Leveraging model memorization for cali-

- brated white-box membership inference. In *29th USENIX Security Symposium*, 2020.
- [5] Ahmed Salem, Yang Zhang, Mathias Humbert, Mario Fritz, and Michael Backes. ML-Leaks: Model and data independent membership inference attacks and defenses on machine learning models. In *Network and Distributed System Security Symposium (NDSS)*, 2019.
 - [6] Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE Symposium on Security and Privacy (SP)*, pages 3–18. IEEE, 2017.
 - [7] Stacey Truex, Ling Liu, Mehmet Ercan Gursoy, Lei Yu, and Wenqi Wei. Demystifying membership inference attacks in machine learning as a service. *IEEE Transactions on Services Computing*, 2019.